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All-Optical Wavelength Converter Concepts for High Data Rate D(Q)PSK Transmission

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ABSTRACT

We investigate the performance of two different all-optical wavelength conversion configurations: four-wave mixing in highly nonlinear fiber and cascaded second harmonic and difference frequency generation in periodically poled Lithium Niobate. Both configurations have the capability to convert phase-modulated signals with high data rates. Error free wavelength conversion of up to 160 Gbit/s DPSK and 320 Gbit/s DQPSK data signals is demonstrated. The converter using highly non-linear fiber can have advantages in network applications in which cascaded wavelength conversion are requested due to its potentially higher conversion efficiency and OSNR. The Lithium Niobate converter generates no phase distortion by wavelength conversion of phase-modulated signals. This could be useful for applications utilizing PSK formats with 2 bit per symbol or more, like DQPSK or 8-PSK.

Keywords: all-optical wavelength conversion, high-speed transmission, phase modulated signals, DPSK, DQPSK

1. INTRODUCTION

All-optical wavelength converters (AOWC) are expected to be essential building blocks in future optical network nodes. AOWCs increase the switching flexibility and provide an enhanced contention resolution of data packet traffic [1]. To achieve these goals, it is desirable that AOWCs operate at high data rates (≥ 100 Gbit/s) and are cascadable. The latter property means that signal degradations by the conversion process have to be small.

Moreover, it is desirable that AOWCs are transparent to the modulation format (amplitude and phase shift keying - ASK and PSK). The latter property is desirable, because it is well known that advanced phase modulation formats, e.g. differential-phase-shift keying (DPSK), or differential quaternary-phase-shift keying (DQPSK), provide higher sensitivity and an increased tolerance against physical system impairments [2]. Therefore, data transmission using these modulation formats has advantages as it increases e.g. the system margin or provides higher bandwidth efficiency. The additional margin can be used to extend transmission distances, to reduce optical power requirements or to relax component specifications. The increased bandwidth efficiency allows to put more traffic on existing wavelength division multiplexing grids.

In this paper, we consider two AOWC configurations. The first is based on four-wave-mixing (FWM) in highly nonlinear fibre (HNLF). We call this configuration HNLF-AOWC in the following. The second AOWC configuration uses a cascaded Second Harmonic and Difference Frequency Generation (cSHG/DFG) scheme in periodically poled $LiNbO_3$ (PPLN). We call this configuration PPLN-AOWC in the following. For both AOWC arrangements, recent experiments demonstrated the operation at high data rates and the operation with different modulation formats: ASK conversion in PPLN [3], differential (quaternary) phase shift keying (D(Q)PSK) conversion in HNLF [4, 5, 6, 7] and PPLN [8, 9].

The paper reports on investigations on both AOWC configurations concerning conversion efficiency, achieved optical signal-to-noise ratio (OSNR), the conservation of phase coherence in the conversion process and the performance in transmission experiments using D(Q)PSK modulation format with data rates of 160 Gbit/s (320 Gbit/s) for symbol rates of 160 Gbaud. Finally, the achieved performances of the two schemes are compared, and their potentials for application are evaluated.

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2. EXPERIMENTAL SET-UP

Figure 1 shows the experimental set-up for the basic characterization of the AOWC and for the wavelength conversion experiments using 160 Gbaud DPSK and DQPSK signals for both the HNLF-AOWC and the PPNL-AOWC.

Both AOWC arrangements comprise two optical paths: the signal and the pump path. The polarisation states and the optical powers of both paths are controlled independently to optimize the conversion process. For the HNLF-AOWC the signal pulses (λ=1553 nm) are injected into an HNLF (length: 1100 m, nonlinear coefficient (γ): 10.3 W^{-1} km^{-1}, zero dispersion wavelength λ_0: 1548 nm, dispersion slope: 0.03 ps/nm^2/km) through the weak arm of a 10 dB coupler, giving a signal power in the HNLF of ~ 3 dBm. An CW pump (ECL) at 1548 nm is phase-modulated with one or two sinusoidal tones by modulation frequencies (f_{mod}) of 73.33 and 110 MHz and phase shifts of 1.0 rad or 1.5 rad to suppress the stimulated Brillouin scattering (SBS). The pump power is coupled into the HNLF (17 dBm) through the strong arm of the 10 dB coupler. A tunable fibre bragg grating (FBG) is used as a notch filter to suppress the high pump power at the output of the HNLF. An optical isolator blocks the reflected power from the FBG. In addition an optical band pass filter (3 dB-width of 5 nm) suppresses the input signal and the residual pump power, allowing only the converted signal at 1542.5 nm to pass.

For the PPLN-AOWC no SBS suppression is needed, resulting in a simplified set-up. The fiber-pigtailed waveguided Ti:PPLN has a length of 93 nm. Its conversion efficiency is determined to be 100 %/W with respect to the launched

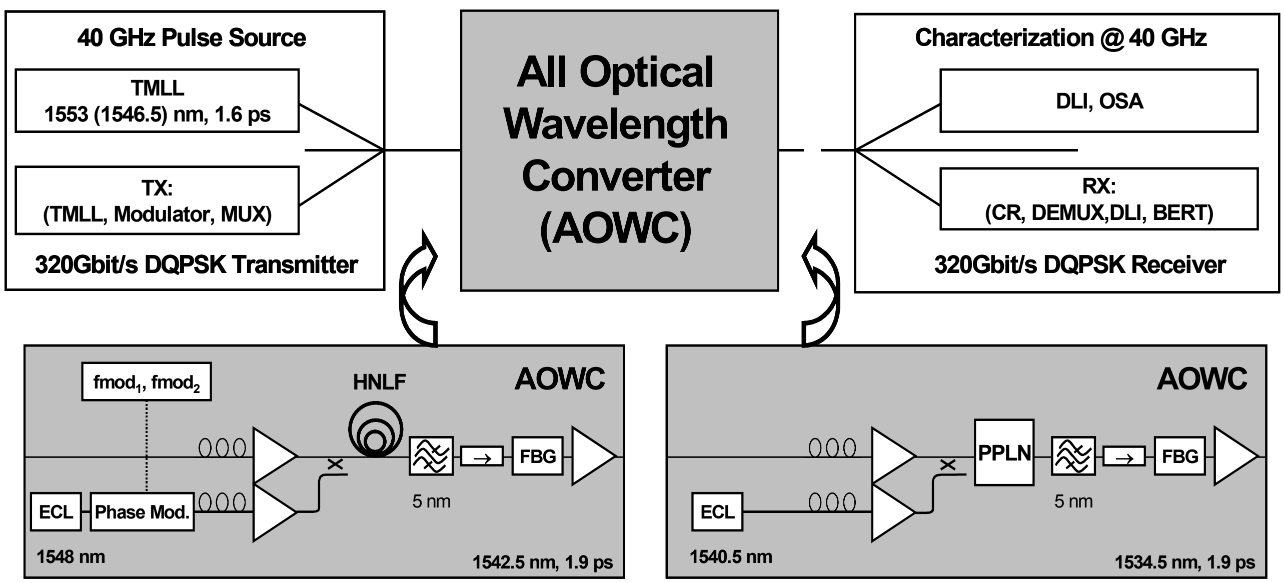

Figure 1: Experimental set up for characterization of the AOWC and for D(Q)PSK wavelength conversion experiments

pump power. For the experiments the pump power in the PPLN is 22.5 dBm at a wavelength of 1540.5 nm. The device is operated at an elevated temperature of 206 °C to avoid photorefractive effects. For the PPLN-AOWC the wavelengths of the signal and the converted signal are 1546.5 nm and 1534.5 nm, respectively. The pump and signal power are suppressed by a filter cascade similar to the one used in the HNLF-AOWC set-up.

For the characterization of the AOWC, a 40 GHz pulse source is used, consisting of a 10 GHz tunable mode-locked laser (TMLL, 1.6 ps pulse width) and a phase-stabilized pulse multiplier. For the wavelength conversion experiments the pulses are phase-modulated in two stages (dual-drive Mach-Zehnder modulator for π phase-shift, and pure phase modulator for $\pi/2$ phase-shift) with electrical PRBS 2^7-1 data signals to create DQPSK modulation format. The modulated pulses are multiplexed (MUX) afterwards up to a symbol rate of 160 Gbaud. Details on the transmitter (TX) and the receiver (RX), which comprised an EAM-based clock recovery, an EAM-based demultiplexer, a delay line interferometer (DLI) and a bit error rate detector (BERT) are presented in detail in reference [6, 9].

The wavelength-converted pulses are analyzed regarding conversion efficiency, optical signal to noise ratio (OSNR) and phase coherence by monitoring the optical spectra (OSA) and the extinction ratio of the two output ports of the DLI.

3. BASIC CHARACTRIZATION

3.1 PPLN-AOWC

Wavelength conversion using Difference Frequency Generation in PPLN is based on the ultra-fast $\chi^{(2)}$ nonlinearity of Lithium Niobate, which is much stronger than the $\chi^{(3)}$ nonlinearity in optical fibers. Therefore, efficient wavelength conversion is possible in very short devices. The DFG wavelength conversion process of data signals in the C-band requires a pump source with a wavelength of about 780 nm. This can be provided by usual telecom equipment, since a 1550 nm light source is internally converted to 780 nm by SHG [10] and the cascaded SHG/DFG is used. The phase matching condition is automatically fulfilled for both SHG and DFG. The following investigations of the characteristics of the PPLN-AOWC were performed:

Conversion Efficiency, Conversion Bandwidth and OSNR:

For the following wavelength conversion experiments, optical pulses (pulse width: 6ps) at the wavelength of 1546.5 nm have been converted to a wavelength of 1534.5 nm. As depicted in Figure 2 (a), a conversion efficiency of -8 dB was achieved by pumping the PPLN with a CW optical power of 22.5 dBm (power within the PPLN). As shown in Figure 1, the complete AOWC is equipped with a filter cascade (band pass filter and FBG) and an EDFA after the non-linear medium to filter out and amplify only the converted signal. The converted output signal of the AOWC is represented in the lower part of Figure 2. It has an OSNR of 37 dB (input OSNR: 51 dB).

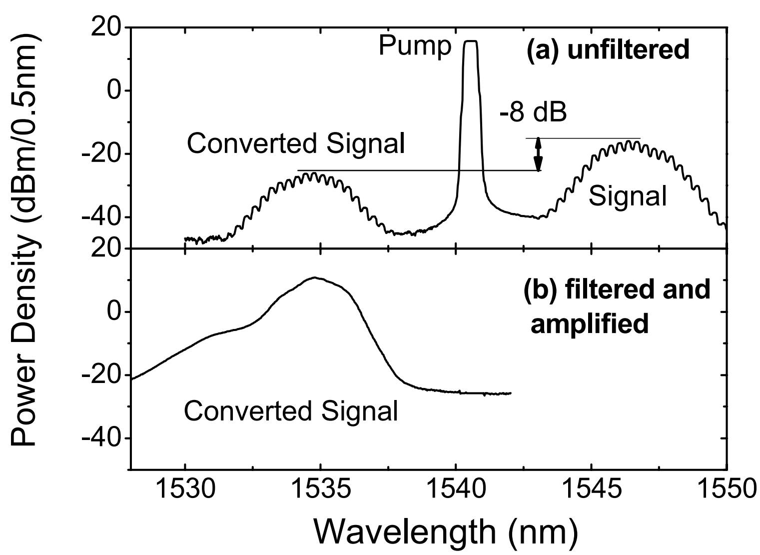

Figure 2: Optical Spectra of PPLN based AOWC, with and without filtering

The conversion bandwidth of the PPLN converter has been investigated by tuning the CW pump within the C-band and by monitoring the optical powers for input and converted signal (Figure 3). A constant conversion efficiency of -8 dB was obtained within the whole C-band due to perfect phase matching for SHG and DFG at the same time. Moreover, Figure 3 features also the attractive capability of the cSHG/DFG configuration to convert a number of WDM channels simultaneously.

On the other hand, more flexibility for the converted signal wavelength can be achieved if the approach of cascaded Sum Frequency and Difference Frequency Generation is applied [3]. In this case, the target wavelength can be chosen independently of the input wavelength by tuning the pump wavelength.

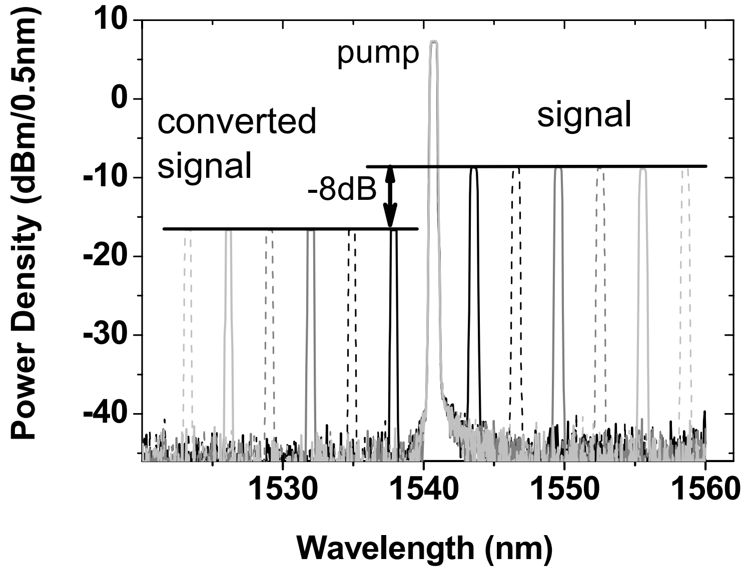

Figure 3: Optical Spectra of PPLN-AOWC to demonstrate the capability of conversion within complete C-band (conversion of tuned CW signal)

Phase coherence:

In transmission systems based on differentially phase modulated signals, high phase coherence between successive bits is required [2]. Phase distortions, e.g. generated by the wavelength conversion process, are not acceptable. Therefore, the wavelength conversion process of the PPLN-AOWC was studied with respect to possible phase distortions.

Phase coherence can be assessed by measuring the extinction ratio (ER) between the output ports of a DLI, if a train of non-modulated bits is fed into the DLI. The recorded ER represents a figure of merit for phase coherence, as it is shown by the following equation:

$$ER(\Phi_{DLI}, \Delta\Phi) = 10 \cdot \log\left\{ \frac{\varepsilon^2 + 1 - 2\varepsilon \cos\left(\Phi_{DLI} + \Delta\Phi\right) + \frac{P_{ASE}}{2P_{sig}}}{\varepsilon^2 + 1 + 2\varepsilon \cos\left(\Phi_{DLI} + \Delta\Phi\right) + \frac{P_{ASE}}{2P_{sig}}} \right\}. \quad (1)$$

According to this equation, the ER is only determined by the phase coherence $\Delta\Phi$ of successive bits (in the experiment, the bit train and hence $\Delta\Phi$ are averaged in observation time), if the power of spontaneous emission P_{ASE} is negligible to signal power P_{sig} or if the OSNR is kept constant. The apparatus parameter ε of the DLI in equation 1 reflects the ratio of the DLI losses for the two optical paths, and has to be determined. The phase of DLI paths (Φ_{DLI}) should be tuneable, to set the DLI to the maximum ER.

In our experiment, the phase coherence of PPLN conversion process was evaluated by comparing the ER of different signals: (a) signal without passing the PPLN, (b) non-converted and (c) converted signal. As shown in Figure 4, the ER were determined for the phase-tuned DLI. For all signals a fixed OSNR of 37 dB was in use.

As Figure 4 shows, very similar maximum ER values were achieved for all signals (bypassed signal: 23.5 dB, non-converted signal: 23 dB and converted signal: 22.5 dB). Obviously no detectable phase distortions are found for the conversion process of the PPLN-AOWC.

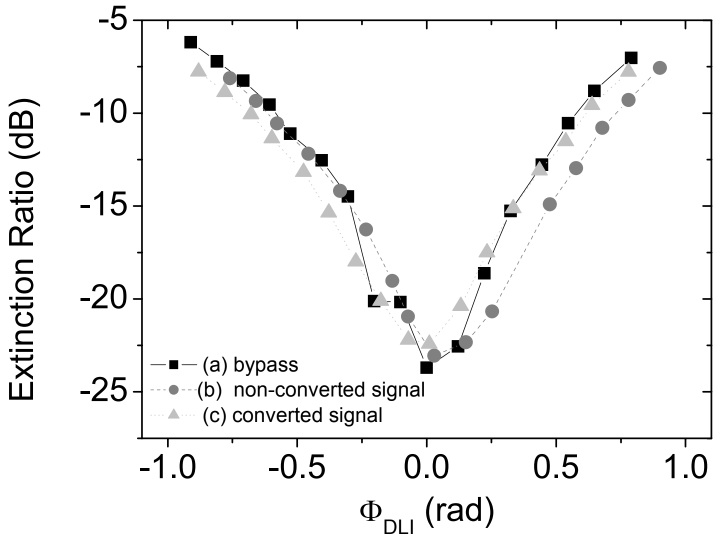

Figure 4: Extinction ratio of DLI output for non-converted, converted PPLN-AOWC signals, and for comparison, for bypassed signal (OSNR of all signals: 37 dB) as a function of DLI phase detuning (Φ_{DLI})

3.2 HNLF-AOWC

Among the different effects of ultra-fast $\chi^{(3)}$ nonlinearity in optical fibers (cross-phase-modulation, self-phase modulation and four-wave mixing), the FWM offers the advantage to be useful for wavelength conversion of both, ASK- and PSK signals [11]. In the simplest configuration, the case of degenerated FWM, an optical input signal and a CW pump wave are fed into the fiber and generate a new light-component which is a phase conjugated replica of the input data signal [12].

To achieve an efficient FWM, the pump power (P_{pump}) and the non-linear coeffiiecnt (γ) of the HNLF have to be large. In addition, the phase mismatch has to be controlled to be low, e.g. by chosing a dispersion flattened highly nonlinear fiber with an extented range of nearly no second order dispersion around λ_0 and low third order dispersion [13].

However, the SBS effect restricts the pump power into the HNLF, and P_{pump} becomes saturated not far from SBS threshold ($\approx$ 10 dBm) [13]. Moreover, at high pump power the generated SBS results in additional noise for amplitude and phase of the converted signal [6, 14, 15]. A well known method to reduce SBS uses phase-modulation of the pump wave, which results in spectral broadening of the pump wave (new spectral side-bands besides the pump carrier) [e.g. 6, 13]. Hence, more technical effort is needed for the HNLF-AOWC than for PPLN-AOWC, and an optimization of the phase-modulated pump scheme has to be performed.

Optimization of phase-modulated pump scheme, phase coherence and OSNR:

The phase modulation of the pump source results in phase modulation of the converted signal itself. For degenerated FWM, the phase of the converted signal (φ_{CS}) depends doubly on the pump phase (φ_{pump}) and singly on the signal phase (φ_S):

$$\varphi_{CS} = 2\varphi_{pump} - \varphi_S \quad . \qquad (2)$$

where the phase of the pump wave is given by, if a harmonic phase modulation is used:

$$\varphi_{pump}(t) = 2\pi f_{pump}\, t + \Delta\Phi_{mod} \cdot \cos(2\pi f_{mod}\, t) \qquad (3)$$

Here f_{pump} is the carrier frequency of the pump source, and $\Delta\Phi_{mod}$ and f_{mod} are the maximum modulation phase shift and the modulation frequency of the phase modulator, respectively.

According equations 2 and 3, the phase distortion of the converted signal goes linearly with $\Delta\Phi_{mod}$, and the phase shift should be chosen as low as possible. On the other hand, the higher the $\Delta\Phi_{mod}$ the larger the number of generated side bands and the more efficient is the spectral broadening of optical pump power and the increase of the SBS threshold. The number of side bands can be increased even for constant $\Delta\Phi_{mod}$, if more than one modulation frequency (single tone) is applied, e.g. by using the dual tone modulation technique.

The single and dual tone pump schemes were compared with respect to:

- FWM efficiency,
- ratio of backscattered SBS power to input pump power (P_{SBS}/P_{pump}),
- phase coherence by detection of DLI's ER,
- OSNR of converted signal (for an input OSNR of 51 dB).

For the investigations the modulation frequencies of 250 MHz (for single and dual tone case) and 166 MHz (as second frequency for dual tone technique) were chosen and the pump power was varied in the range of 11 to 20 dBm (coupled into the HNLF). The phase shift was varied too, in the range of 1 to 2.5 rad:

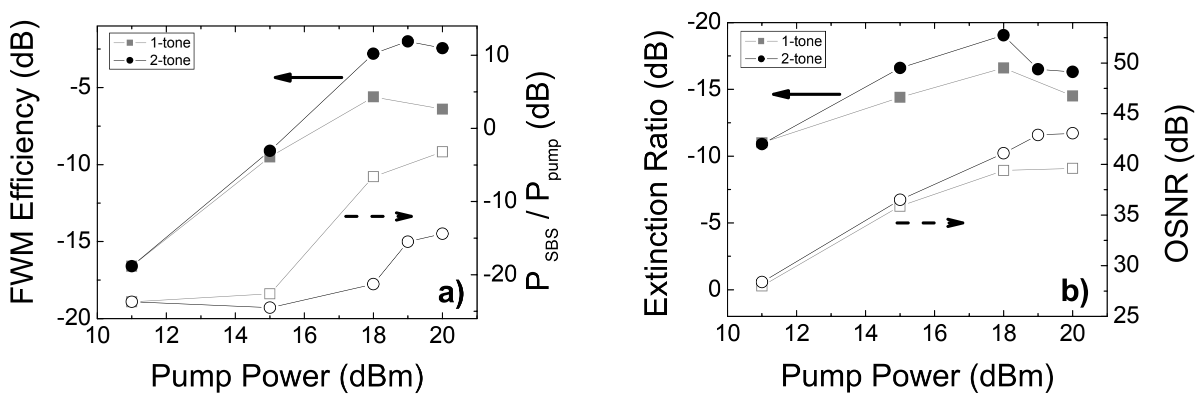

Figure 5: Comparison of single and dual tone pump scheme: a): FWM efficiency and ratio of backscattered SBS and input pump power (P_{SBS}/P_{pump}), b): ER of DLI and OSNR of converted signal (input OSNR of 51 dB)

As shown in Figure 5 (a), the dual tone scheme provides a higher FWM efficiency, because the ratio P_{SBS}/P_{pump} can be controlled to be moderate, even with high pump power. For example, an FWM efficiency of −3 dB for a pump power of 18 dBm, and an acceptable P_{SBS}/P_{pump} below −20 dB (shown in Figure 5 a) can be obtained at the same time. For single tone technique, only a pump power of 15 dBm can be used, if a limited backscattered power below −20 dB is accepted. Hence, only a conversion efficiency of −10 dB can be achieved by this technique. The lower FWM efficiency results in an OSNR penalty of −5 dB compared to dual tone technique (Figure 5 b: comparison of OSNR for single and dual tone technique for 15 and 18 dBm pump power, respectively). Finally, the phase coherence has been investigated for the two different pump schemes by means the ER of DLI. In Figure 5 (b), we compare the ER's for pump powers of 15dB and 18 dB for the single and dual tone scheme respectively. It is indicated a lower phase distortion for the dual tone technique, namely an ER value of −19.5 dB. This is an improvement of 5 dB compared to the single tone technique (Figure 5 b).

In summary, the dual tone pump scheme provides a more effective SBS suppression than the single tone technique, even with higher pump power and better OSNR for the converted signal.

Equation 2 and 3 show, that the occurrence of phase errors grows directly with the modulation frequency (f_{mod}). Therefore, the modulation frequency should be kept as low as possible. However, the minimum applicable frequency is determined by the Brillouin gain bandwidth ($\Delta\nu_B$). The side bands, generated by phase modulation, have to be outside $\Delta\nu_B$ to avoid SBS [16].

The minimum acceptable frequency has been obtained experimentally by monitoring the SBS power (SBS-backscattered CW pump power) vs. phase modulation frequency, as shown in Figure 6. The backscattered CW power decreases with increasing modulation frequencies up to 70 MHz and is constant for f_{mod}.> 70 MHz, which in agreemant with the the

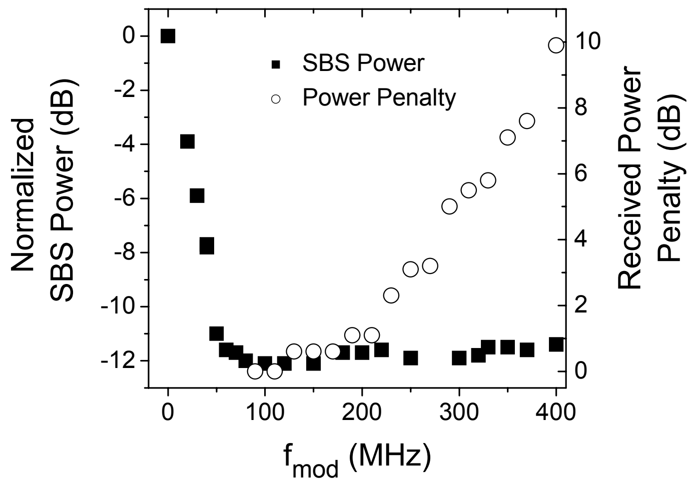

Figure 6: Normalized SBS power and received power penalty (BER of 10^{-9} of 80 Gbit/s DQPSK converted signal) vs. phase modulation frequency f_{mod}.

SBS gain bandwidth $\Delta\nu_B$ [12, 16].

The maximum acceptable frequency was determined by measurements of the received power for a BER of 10^{-9} of a 80 Gbit/s DQPSK signal, which has been converted by the HNLF-AOWC. Figure 6 represents the power penalty vs. modulation frequency. It indicates that the frequency should be below 150 MHz in order to keep an inevitable power penalty below 1 dB.

In conclusion, the modulation frequencies have to be chosen within a narrow frequency band between 70 and 150 MHz. In addition, the dual tone technique is favored due to improved FWM efficiency and OSNR of converted signal, even with lower phase distortion.

Conversion Efficiency, Conversion Bandwidth and OSNR:

For all following conversion experiments, the dual tone technique with modulation frequencies of 73.3 MHz and 110 MHz and with phase shifts of 1.0 rad and 1.5 rad, respectively, were chosen.

In the conversion experiments, the phase modulated RZ signal with a wavelength of 1553 nm and pulse width of 1.6 ps have been converted to a wavelength of 1543 nm, as shown in Figure 7. The conversion efficiency was found to be -5 dB since a pump power of 18 dBm (within the HNLF) was applied. The OSNR of the AOWC output signal, which has been spectrally filtered and post-amplified, was determined to be 41 dB.

For evaluation of HNLF-AOWC's conversion bandwidth, the input signal and the pump wavelength have been tuned within the C-band and the power levels of the converted signal were determined, also by monitoring the FWM spectra as shown in Figure 8. For the three different pump wavelengths of 1542nm, 1548nm and 1552 nm, the achieved conversion efficiencies, are presented, wherein the input signal wavelength was tuned in the range of 1533nm to 1566 nm.

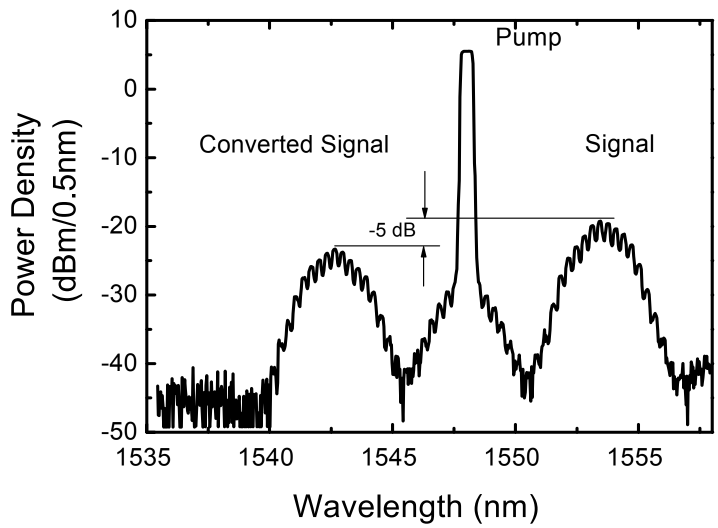

Figure 7: Optical spectrum of FWM in HNLF-AOWC to demonstrate the conversion efficiency

Figure 8 shows, that wavelength conversion can be performed within the complete C-band, if a dispersion-flattened HNLF is used. However, for this fibre, a drop down of conversion efficiency by about – 5dB with respect to the maximum efficiency has to be accepted.

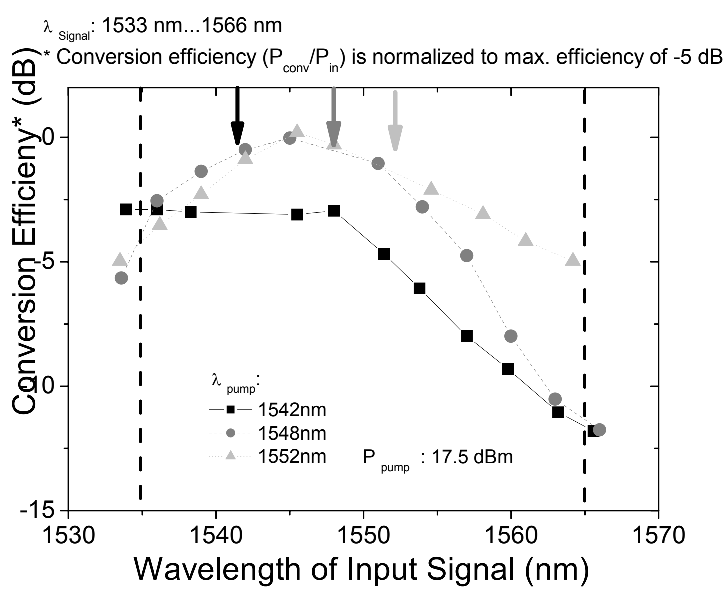

Figure 8: Conversion efficiency of RZ signals (normalized to maximum FWM conversion efficiency of −5 dB) vs. input signal wavelength for three different pump wavelengths (1542, 1548 and 1552 nm)

4. CONVERSION EXPERIMENTS

4.1 PPLN-AOWC

The performance of the PPLN based AOWC to convert phase encoded signals has been investigated by bit error ratio (BER) conversion experiments using DPSK and DQPSK RZ signals with symbol rates of 40, 80 and 160 Gbaud. The set-up for these experiments is shown and described in chapter 2.

Figure 9 shows BER measurements versus the received average power at the 160 Gbaud receiver. The curves are plotted for back-to-back measurements (b2b, without AOWC) and for measurements of the converted signal (conv., with AOWC). The wavelength conversion of DPSK data (Figure 9 a) caused a negligible penalty in receiver sensitivity (less than 0.5 dB at BER 10^{-9}) for 40, 80 and 160 Gbit/s compared to b2b.

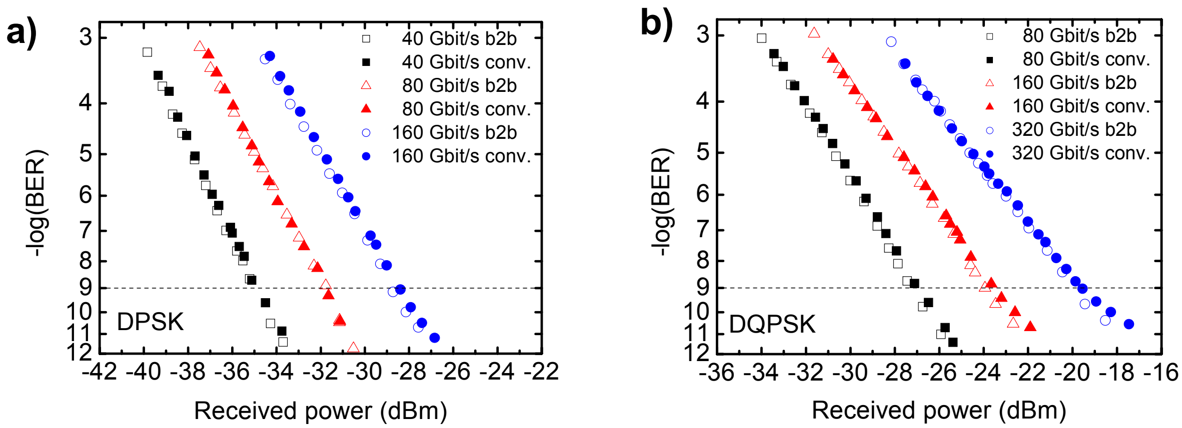

Figure 9: PPLN-AOWC-BER measurements for 40, 80, and 160 Gbit/s DPSK (a), and for 80, 160, and 320 Gbit/s DQPSK (b)

For DQPSK an error free operation with negligible penalty (lower than 0.5 dB for a BER of 10^{-9}) was achieved as well, but in this case for 80, 160 and 320 Gbit/s (Figure 9 b).

The BER measurements showed also, that the PPLN based AOWC did not disturb the phase of the data signals. Moreover, the conversion performance was not limited by the bit rate up to 160 Gbaud.

4.2 HNLF-AOWC

The performance of the HLNF-AOWC based on dual tone scheme has been investigated by BER conversion experiments too. Again, DPSK and DQPSK modulation formats with symbol rates of 40, 80 and 160 Gbaud/s were used.

In Figure 10 BER measurements without wavelength conversion (b2b) and with wavelength converter (conv) are shown as a function of the average receiver power. The performance for conversion of DPSK data is shown in Figure 10 a for 40, 80 and 160 Gbit/s. The wavelength conversion is seen to cause a negligible penalty in receiver sensitivity (less than 0.2 dB at BER 10^{-9}).

For DQPSK wavelength conversion, the BER measurements for data rates of 80, 160 and 320 Gbit/s are shown in Figure 10 b. Error free operation was achieved, however, with a 2-3 dB penalty independently of the data rate. The penalty for the converted DQPSK signals is due to the not negligible phase distortion, which was inserted by the phase modulation, according to the findings in previous section. For the single tone scheme, a minimum penalty of only 4 dB can be achieved [6].

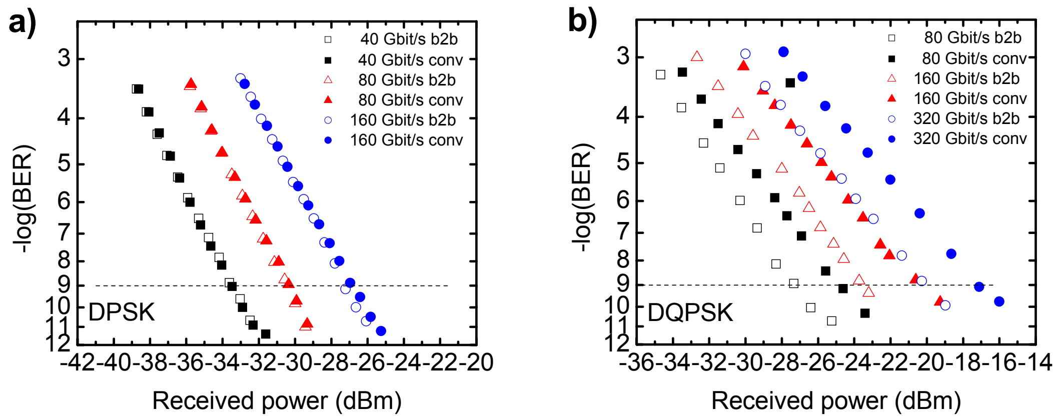

Figure 10: HNLF-AOWC-BER measurements for 40, 80, and 160 Gbit/s DPSK (a), and for 80, 160, and 320 Gbit/s DQPSK (b)

Finally, it turns out, that the performance of the HNLF-AOWC is not limited by the bit rate too.

5. SUMMARY

The realization of the HNLF-AOWC and the PPLN-AOWC requires similar technical effort, as outlined in chapter 2. In both cases, a strong optical CW pump (18 -23 dBm) has to be coupled into the AOWC-configuration in order to accomplish the mixing process in the nonlinear medium. Additionally, a tunable and efficient optical filter configuration is needed in the final stage of the AOWC in order to suppress the high pump power and to allow only the converted signal to pass. A post-amplification of the converted signal is required. For the HNLF-AOWC, the optical pump path needs additional effort to suppress the SBS, e.g. a well adjusted phase-modulated pump source.

The performance of the investigated PPLN-AOWC and HNLF-AOWC is concisely compared in Table 1.

The PPLN-AOWC provides a superior phase coherence, as pointed out by the high extinction ratio of the DLI with ER ≈ 22.5 dB, which is close to the ER without AOWC (≈ 23 dB). The PPLN converter shows a extended conversion bandwidth (the complete C-Band is covered), with a constant conversion efficiency of – 8 dB. The moderate conversion efficiency limits the OSNR of the converted signal, which was found to be 37 dB, since the OSNR of the input signal was 51 dB.

Table 1: Overview on wavelength conversion performance of PPLN- and HNLF-AOWC, *: OSNR of converted signal for input OSNR of 51 dB, +: extinction ratio of DLI for train of converted pulses (ER of non-converted pulses: 23 dB)

AOWC	Conversion Efficiency (dB)	Conversion Bandwidth (nm)	OSNR* (dB)	ER+ / Phase Coherence (dB)	Error Free Conversion (160 Gbaud-BER: 10^{-9})	Received Power Penalty @ 160 Gbaud-BER of 10^{-9} (dB)
PPLN	-8	Complete C-band	37	22.5	DPSK: yes DQPSK: yes	DPSK: < 0.5 DQPSK: < 0.5
HNLF	-5	Complete C-band (5dB penalty of conversion efficiency within C-band)	41	19.5	DPSK: yes DQPSK: yes	DPSK: < 0.4 DQPSK: 2-3

Compared to the PPLN-AOWC, the HNLF-AOWC shows higher conversion efficiency (- 5 dB). The OSNR of the converted signal is found to be 41 dB. However, the conversion efficiency drops down, if the signal wavelength is tuned away from zero dispersion wavelength. The conversion efficiency declines by 5 dB within the C-Band. The phase-modulated pump scheme generates phase distortions, which are confirmed by a degraded extinction ratio of the DLI with ER ≈ 19.5 dB. The pump scheme of HNLF-AOWC has the potential to allow higher pump power and even improved conversion efficiency.

The wavelength conversion of DPSK data by both AOWC configurations was found to be error free and causes a negligible penalty in receiver sensitivity (less than 0.5 dB at BER 10^{-9}) for 40, 80 and 160 Gbit/s compared to back to back. For DQPSK an error free operation with negligible penalty was achieved for 80, 160 and 320 Gbit/s as well for the PPLN-AOWC. The conversion experiments show, that the AOWC with PPLN does not disturb the phase of the data signals and that its performance is not limited by the bit rate up to 160 Gbaud.

For the HNLF-AOWC, the DQPSK conversion was in fact error free, however a penalty of 2-3 dB in receiver sensitivity has to be accepted. This penalty is due to the phase modulation of the CW pump resulting in reduced phase coherence for the converted signal, which was confirmed by phase coherence investigations too.

The investigations of the HNLF-AOWC and the PPLN-AOWC indicate the feasibility of both configurations for high-speed wavelength conversion of phase modulated data signals in future photonic networks.

The HNLF-AOWC can have advantages in network applications containing cascaded wavelength conversion due to its potentially higher conversion efficiency and OSNR. The PPLN-AOWC shows the better performance for phase-modulated signals with negligible penalty up to 320 Gbit/s. This could be an advantage for network applications using PSK formats with 2 bit per symbol or more, like DQPSK or 8-PSK.

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